INCH-POUND

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MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, DELAY, M30A1, M32A1, M43A1, M92, AND M120 METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M30A1, M32A1, M43A1, M92, and M120 Cartridge Actuated Initiator Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550 Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109 Quality Assurance Terms And Definitions
MIL-STD-413 Visual Inspection Guide for Elastomeric O-rings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 3730), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

10520225	Initiator, Cartridge Actuated, Delay, Metal Parts Assembly
8798143	Box, Fiberboard (Inner), Packing, Ammunition, For Initiator, Cartridge Actuated, Delay, M10, M30A1, M31, M32A1, M42, M43A1, M72, M92, or M114
8798142	Box, Packing, Ammunition, For Initiator, Cartridge Actuated, Delay, M10, M30A1, M31, M32A1, M42, M43A1, M72, M92, or M114
9216933	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Initiator, Cartridge Actuated, Delay, M10, M30A1, M31, M32A1, M42, M43A1, M72, M92, or M114

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Data Control Branch (Code 3720), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to the first article inspection (see 4.4 and 6.3).

3.2 Materials. All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

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3.2.1 Raw material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.3 Hardness and other physical properties. Hardness and other physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.4 Hydrostatic pressure. The initiator chamber and cap shall withstand a hydrostatic pressure of 5500 to 6500 pounds per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure (see 6.4.1).

3.5 Firing pin release. The firing pin of the metal parts assembly, less safety pin, cartridge retainer and chamber, shall release from the cocked position when the application of an axial load of 20 to 35 pounds has withdrawn the initiator pin 3/4 inch, maximum (see 4.5.3.3).

3.6 Initiator pin security. Application of an axial load of 200 to 220 pounds applied to the initiator pin of the metal parts assembly shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin (see 4.5.3.4).

3.7 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

3.8 Protective finish. Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration.

3.9 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using procedures in 4.6.6.

3.10 Workmanship. Initiators shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.10.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.10.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.10.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77°F)
b. Altitude: sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

4.4 First article inspection. At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of five metal parts assemblies, nine sets of unassembled parts, and one set of unassembled parts without plating or final protective finish. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After

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inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new first article samples until an acceptable sample is submitted (see 6.3).

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.5.1.2 Size of lot. For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 assemblies, maximum, unless otherwise specified by the contract.

4.5.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.5.2.1 Retainer, Cartridge, Drawing 8594860, covering a detail of Drawing 10520225.

Characteristics	Defect	Method of Inspection
Critical		
1	Metering slots in base missing	Visual
Major		
101	Pitch diameter of thread, max	Gage
102	Minor diameter of thread, max	Gage
103	Concentricity of cavity with pitch diameter of thread	Gage
104	Concentricity of counterbore with cavity	Gage
105	Perpendicularity of pitch diameter of thread with base	Gage
106	Symmetry of metering slots in base with body	Gage
107	Diameter of cavity, max	Gage
108	Depth of cavity	Gage
109	Depth to bottom of counterbore	Gage
110	Depth of metering slots in base	Gage
111	Length of effective thread, min	Gage
112	Length of thread, max	Gage
113	Internal radius at mouth cavity	Gage

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114	Total length	Measure	
115	Width of metering slots in base		Measure
116	Drill point in base, max	Measure	
117	Number of elongated slots incorrect		Visual
118	Thread damaged or not full	Visual	
119	Metal defective	Visual	
120	Protective finish inadequate or defective		Visual
121	Surface finish on internal radius and base improper		Visual
122	Burr	Visual	
123	Foreign matter	Visual	
Minor			
201	Pitch diameter of thread, min		Gage
202	Diameter of cavity, min	Gage	
203	Diameter of counterbore	Gage	
204	Diameter of head	Gage	
205	Depth of counterbore		Gage
206	Diameter of body	Measure	
207	Length of elongated slots	Measure	
208	Length of head	Measure	
209	Width of elongated slots	Measure	
210	Location of elongated slots		Measure
211	Angle between head and body	Measure	
212	Angular location of metering slots		Visual
213	Marking missing, incorrect, or illegible		Visual
214	Profile of elongated slots	Visual	
215	Radius missing or incorrect		Visual
216	Surface finish improper, except as otherwise classified		Visual

4.5.2.2 Chamber, Initiator, Drawing 8594857, covering a detail of Drawing 10520225.

Critical		
1	Absence of output port	Visual
Major		
101	Pitch diameter of large thread, min	Gage
102	Major diameter of large thread, min	Gage
103	Pitch diameter of small thread, min	Gage
104	Major diameter of small thread, min	Gage
105	Concentricity of counterbore with pitch diameter of large thread	Gage
106	Concentricity of datum diameter with small thread	Gage
107	Diameter of counterbore	Gage
108	Depth of cavity	Gage
109	Thickness of wall, min	Gage

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110	Location of datum diameter	Gage	
111	Profile of gas port	Gage	
112	Thread damaged or not full	Visual	
113	Chamfer at base of cavity missing or incorrect		Visual
114	Metal defective	Visual	
115	Protective finish inadequate or defective		Visual
116	Surface finish at end of gas port improper		Visual
117	Burr	Visual	
118	Foreign matter	Visual	

Minor

201	Pitch diameter of large thread, max		Gage
202	Pitch diameter of small thread, max		Gage
203	Diameter of cavity	Gage	
204	Diameter of hole	Gage	
205	Diameter of small thread pilot		Gage
206	Diameter of large thread pilot		Gage
207	Length of large thread pilot		Gage
208	Length of gas port	Gage	
209	Thickness of wall, max	Gage	
210	Location of small thread	Gage	
211	Depth of counterbore	Measure	
212	Total length	Measure	
213	Length to end of large thread		Measure
214	Marking missing, incorrect, or illegible		Visual
215	Radius or chamfer missing or incorrect, except as otherwise classified		Visual
216	Surface finish improper, except as otherwise classified		Visual

4.5.2.3 Pin, Initiator, Drawing 8595403, covering a detail of Drawing 10520225.

Critical
None defined

Major

101	Concentricity of groove in head with body		Gage
102	Concentricity of head with body		Gage
103	Diameter of body	Gage	
104	Diameter of groove in body	Gage	
105	Diameter of groove in head, min		Gage
106	Length to shoulder at head, min		Gage
107	Location of groove in head	Gage	
108	Location of groove in body	Gage	
109	Profile of groove in body	Measure	
110	Profile of groove in head	Measure	

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111	Metal defective	Visual
112	Protective finish inadequate or defective	Visual
113	Surface finish on body and body groove improper	Visual
114	Burr	Visual
115	Foreign matter	Visual

Minor

201	Diameter of groove in head, max	Gage
202	Diameter of pull hole	Gage
203	Diameter of undercut, min (if applicable)	Gage
204	Length to shoulder at head, max	Gage
205	Width of head	Gage
206	Width of shoulder at body groove	Gage
207	Width of groove in head	Gage
208	Thickness of head	Gage
209	Total length	Measure
210	Width of undercut, max (if applicable)	Measure
211	Location of pull hole	Measure
212	Angle at base of body	Measure
213	Profile of head	Measure
214	Marking missing, incorrect, or illegible	Visual
215	Radius or chamfer missing or incorrect	Visual
216	Surface finish improper, except as otherwise classified	Visual

4.5.2.4 Cap, Drawing 8595404, covering a detail of Drawing 10520225.

Critical
None defined

Major

101	Pitch diameter of large thread, max	Gage
102	Minor diameter of large thread, max	Gage
103	Pitch diameter of small thread, max	Gage
104	Minor diameter of small thread, max	Gage
105	Concentricity of pitch diameter of large thread with pitch diameter of small thread, including perpendicularity with base of large thread cavity	Gage
106	Concentricity of hole counterbore with hole	Gage
107	Concentricity of hole with pitch diameter of small thread, including perpendicularity with base of large thread cavity	Gage
108	Symmetry of flats with pitch diameter, large thread	Gage
109	Diameter of hole counterbore	Gage
110	Diameter of pin holes	Gage
111	Depth of large thread cavity	Gage
112	Depth of small thread cavity	Gage

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113	Length of effective large thread, min	Gage
114	Length of effective small thread, min	Gage
115	Height, max	Gage
116	Thickness of wall, min	Measure
117	Internal radius at base of large thread cavity missing or incorrect	Visual
118	Thread damaged or not full	Visual
119	Metal defective	Visual
120	Protective finish inadequate or defective	Visual
121	Surface finish of hole improper	Visual
122	Burr	Visual
123	Foreign matter	Visual

Minor

201	Pitch diameter of large thread, min	Gage
202	Pitch diameter of small thread, min	Gage
203	Pitch diameter of setscrew thread	Gage
204	Minor diameter of setscrew thread, max	Gage
205	Diameter of boss	Gage
206	Diameter of flange holes	Gage
207	Depth of hole counterbore	Gage
208	Total length	Gage
209	Length between flange holes	Gage
210	Width across body flats	Gage
211	Width of groove	Gage
212	Location of pin holes	Gage
213	Location of groove	Gage
214	Diameter of groove	Measure
215	Length of flange	Measure
216	Width of flange	Measure
217	Thickness of flange	Measure
218	Location of flange hole	Measure
219	Location of setscrew holes	Measure
220	Location of initiator pin hole	Measure
221	Marking missing, incorrect, or illegible	Visual
222	Radius or chamfer missing or incorrect, except as otherwise classified	Visual
223	Surface finish improper, except as otherwise classified	Visual

4.5.2.5 Pin, Safety, Drawing 8595405, covering a detail of Drawing 10520225.

Critical
None defined

Major

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101	Diameter of wire	Measure	
102	Profile incorrect	Visual	
103	Metal defective	Visual	
104	Protective finish inadequate or defective		Visual
105	Burr or sharp edges	Visual	
106	Foreign matter	Visual	

Minor

201	Diameter of head	Measure	
202	Total length	Measure	

4.5.2.6 Spring, Initiator, Drawing 8595406, covering a detail of Drawing 10520225.

Critical
None defined

Major

101	Load at specified heights	Gage	
102	Height solid, max	Gage	
103	Ends incorrect	Visual	
104	Protective finish inadequate or defective		Visual
105	Burr or sharp edges	Visual	
106	Foreign matter	Visual	

Minor

201	Diameter outside, closed, max		Gage
202	Diameter inside, min	Gage	
203	Free length	Gage	

4.5.2.7 Pin, Firing, Drawing 8595407, covering a detail of Drawing 10520225.

Critical
None defined

Major

101	Concentricity of cavity with body		Gage
102	Concentricity of striker with body		Gage
103	Concentricity of shoulder with body		Gage

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104	Diameter of body	Gage
105	Diameter of cavity	Gage
106	Diameter of striker	Gage
107	Diameter of ball hole, max	Gage
108	Depth of ball hole counterbore, max	Gage
109	Length from striker to body	Gage
110	Radial profile of striker	Gage
111	Radial location of ball holes	Visual
112	Metal defective	Visual
113	Protective finish inadequate or defective	Visual
114	Surface finish on body and striker improper	Visual
115	Burr	Visual
116	Foreign matter	Visual

Minor

201	Diameter of cavity counterbore	Gage
202	Diameter of ball hole, min	Gage
203	Diameter of ball hole counterbore	Gage
204	Diameter of shoulder	Gage
205	Depth of cavity counterbore, min	Gage
206	Depth of cavity	Gage
207	Depth of ball hole counterbore, min	Gage
208	Total length	Gage
209	Length of striker	Gage
210	Length to ball hole	Gage
211	Diameter of drill point in cavity, max	Measure
212	Marking missing, incorrect, or illegible	Visual
213	Radius or chamfer missing or incorrect, except as otherwise classified	Visual
214	Surface finish improper, except as otherwise classified	Visual

4.5.2.8 Housing, Firing Pin, Drawing 8595408, covering a detail of Drawing 10520225.

Critical
None defined

Major

101	Pitch diameter of small thread, min	Gage
102	Major diameter of small thread, min	Gage
103	Pitch diameter of large thread, min	Gage
104	Major diameter of large thread, min	Gage
105	Concentricity of counterbore with cavity	Gage
106	Concentricity of flange with pitch diameter of small thread	Gage
107	Concentricity of groove with flange	Gage
108	Concentricity of striker hole with cavity	Gage

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109	Concentricity of pitch diameter of large thread with cavity		Gage
110	Concentricity of cavity with pitch diameter of small thread, including perpendicularity with face of flange		Gage
111	Concentricity of boss cavity with cavity		Gage
112	Perpendicularity of boss face with pitch diameter of large thread		Gage
113	Diameter of cavity, max	Gage	
114	Diameter of datum location	Gage	
115	Diameter of groove	Gage	
116	Diameter of flange, min	Gage	
117	Length from rear flange to cavity		Gage
118	Width of rear flange	Gage	
119	Angle at datum diameter	Gage	
120	Height of boss	Gage	
121	Diameter of small thread undercut, min		Measure
122	Profile of groove	Measure	
123	Radius at root of boss	Measure	
124	Thread damaged or not full	Visual	
125	Metal defective	Visual	
126	Protective finish inadequate or defective		Visual
127	Surface finish of cavity improper		Visual
128	Burr or sharp corners	Visual	
129	Foreign matter	Visual	
Minor			
201	Pitch diameter of small thread, max		Gage
202	Pitch diameter of large thread, max		Gage
203	Diameter of cavity, min	Gage	
204	Diameter of boss cavity	Gage	
205	Diameter of flange, max	Gage	
206	Diameter of body	Gage	
207	Diameter of counterbore	Gage	
208	Diameter of undercut in cavity, max (if applicable)		Gage
209	Diameter of striker hole	Gage	
210	Diameter of small thread pilot		Gage
211	Depth to bottom of cavity	Gage	
212	Thickness at striker hole	Gage	
213	Total length	Gage	
214	Length from small thread pilot to flange		Gage
215	Length of flange	Gage	
216	Length of body	Gage	
217	Location of flange	Gage	
218	Width of undercut in cavity (if applicable)		Measure
219	Width of undercut, small thread		Measure
220	Width of groove	Measure	

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221	Variation at groove sidewalls, max	Measure
222	Marking missing, incorrect, or illegible	Visual
223	Radius or chamfer missing or incorrect, except as otherwise classified	Visual
224	Surface finish improper, except as otherwise classified	Visual

4.5.2.9 Seal, Drawing 10524102, covering a detail of Drawing 10520225.

Critical
None defined

Major

101	Diameter outside, min	Gage
102	Width, min	Gage
103	Thickness, min	Gage
104	Cut, cracked, or split	Visual
105	Flash	Visual
106	Foreign matter	Visual

Minor

201	Diameter outside, max	Gage
202	Width, max	Gage
203	Thickness, max	Gage

4.5.2.10 Assembly, less Initiator Pin, Safety Pin, Chamber, Cartridge Retainer, Protective Cap, Drawing 10520225, less Drawings 8595403, 8595405, 8594857, 8594860 and MIL-C-5501/17.

Critical

1	Protrusion of firing pin	Gage
2	Steel ball missing	Gage

Major
None defined

Minor
None defined

4.5.2.11 Assembly, Drawing 10520225.

Critical
None defined

Major

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101	Initiator pin not free to rotate	Manual
102	Safety pin loose or deformed	Manual/Visual
103	Any part missing, except as otherwise classified	Visual
104	Improper assembly	Visual
105	Thread damaged or not full	Visual
106	Marking missing, incorrect, or illegible	Visual
107	Burr	Visual
108	Foreign matter	Visual

Minor
None defined

4.5.2.12 Box, Fiberboard (Inner) Packing Ammunition, Drawing 8798143.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual
202	Box damaged	Visual
203	Marking missing, incorrect, or illegible	Visual
204	Cut, damaged, or improperly sealed bag	Visual

4.5.2.13 Box, Packing, Ammunition, Drawing 8798142.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Visual/Manual
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.5.2.14 Box, Fiberboard (Outer) Alternative, Packing Ammunition, Drawing 9216933.

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Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual/Visual	
202	Carton broken or damaged	Visual	
203	Number of units contained incorrect		Visual
204	Marking missing, incorrect, or illegible		Visual
205	Tape incomplete	Visual	

4.5.3 Inspection testing. The following tests shall be performed in accordance with 4.6.

4.5.3.1 Hardness. Fifty units from each heat treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.5.3.2 Hydrostatic pressure. Each chamber and cap shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with 3.4 shall be cause for rejection of the subassembly.

4.5.3.3 Firing pin release. The initiator pin of each subassembly shall be subjected to the firing pin release test. Failure of any subassembly to comply with the requirements of 3.5 shall be cause for rejection of the subassembly.

4.5.3.4 Initiator pin security. The initiator pin of each subassembly shall be subjected to the initiator pin security test. Failure of any subassembly to comply with the requirements of 3.6 shall be cause for rejection of the subassembly.

4.5.3.5 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.5.3.6 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.6.6. Failure of any unit in the inspection sample to comply with 3.9 shall be cause for rejection of the assembly.

4.6 Test methods and procedures.

4.6.1 Hardness. Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

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4.6.2 Hydrostatic pressure. Prior to application of protective finish the cap and chamber shall be mounted in the device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.4.

4.6.3 Firing pin release. The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the device which registers the amount of pull.

4.6.4 Initiator pin security. The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull. The test shall be performed with the safety pin in each position.

4.6.5 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.6.6 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.7 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8798143 and 8798142. Level B shall be in accordance with Drawings 8798143 and 9216933.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

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b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. If first article inspection is required (see 3.1 and 6.3)

d. The responsible inspection activity (see 4.1)

e. Any special inspection requirements (see 4.3)

f. Level of packing and marking (see 5.1)

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors that the items should be a first article sample. The first article should consist of 5 metal parts assemblies, 9 sets of unassembled parts, 1 set of unassembled parts without plating or final protective finish. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definition.

6.4.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

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Custodian:

Preparing activity
Navy - OS

Navy - OS

Air Force - 99

(Project 1377-0D99)

Review activity:
Air Force - 70

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